glacier_exploratory

Load packages

```
pacman::p_load('tidyverse',
               'here',
               'janitor',
               'ggthemes',
               'terra',
               'tmap',
               'patchwork',
               'stars',
               'rnaturalearth'
               )
```

Read data

```
glacier <- st_read(here('data','LIA_glacier_outlines_Alps', 'LIA_Alps_Reinthaler_Paul_202
```

```
Reading layer `LIA_Alps_Reinthaler_Paul_2024' from data source
  `/Users/petervitale/Desktop/MEDS/EDS-240/eds240-
infographic/data/LIA_glacier_outlines_Alps/LIA_Alps_Reinthaler_Paul_2024.shp'
  using driver `ESRI Shapefile'
Simple feature collection with 4448 features and 10 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: 258280.9 ymin: 4885573 xmax: 873103.5 ymax: 5276949
Projected CRS: WGS 84 / UTM zone 32N
```

```
lake_30 <- read_delim(here('data','VianiC_2018',
                           'datasets', '1930s_glaciallakes.tab')) %>%
  clean_names()
```

```
Rows: 43 Columns: 8
── Column specification ────────────────────────────────────────────────────────
Delimiter: "\t"
chr (2): Event, Mountain gr
dbl (6): Station, Date/Time, Latitude, Longitude, Area [m**2], Perimeter [m]

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
lake_70 <- read_delim(here('data','VianiC_2018',
                           'datasets', '1970s_glaciallakes.tab'))%>%
  clean_names()
```

```
Rows: 66 Columns: 8
── Column specification ────────────────────────────────────────────────────────
Delimiter: "\t"
chr (2): Event, Mountain gr
dbl (6): Station, Date/Time, Latitude, Longitude, Area [m**2], Perimeter [m]

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
lake_80 <- read_delim(here('data','VianiC_2018',
                           'datasets', '1980s_glaciallakes.tab'))%>%
  clean_names()
```

```
Rows: 133 Columns: 8
── Column specification ────────────────────────────────────────────────────────
Delimiter: "\t"
chr (2): Event, Mountain gr
dbl (6): Station, Date/Time, Latitude, Longitude, Area [m**2], Perimeter [m]

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
lake_90 <- read_delim(here('data','VianiC_2018',
                           'datasets', '1990s_glaciallakes.tab'))%>%
  clean_names()
```

```
Rows: 178 Columns: 8
── Column specification ────────────────────────────────────────────────────────
Delimiter: "\t"
chr (2): Event, Mountain gr
dbl (6): Station, Date/Time, Latitude, Longitude, Area [m**2], Perimeter [m]

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
lake_06 <- read_delim(here('data','VianiC_2018',
                           'datasets', '2006_glaciallakes.tab'))%>%
  clean_names()
```

```
Rows: 214 Columns: 8
── Column specification ────────────────────────────────────────────────────────
Delimiter: "\t"
chr (2): Event, Mountain gr
dbl (6): Station, Date/Time, Latitude, Longitude, Area [m**2], Perimeter [m]

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
lake_12 <- read_delim(here('data','VianiC_2018',
                           'datasets', '2012_glaciallakes.tab'))%>%
  clean_names()
```

```
Rows: 186 Columns: 8
── Column specification ────────────────────────────────────────────────────────
Delimiter: "\t"
chr (2): Event, Mountain gr
dbl (6): Station, Date/Time, Latitude, Longitude, Area [m**2], Perimeter [m]

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Join the `lake` datasets into one - all have the same column names so we can `bind_rows()`

```
lake_joined <- bind_rows(list(lake_30, lake_70, lake_80,
                           lake_06, lake_12))
```

I want to explore these a bit more so lets find the unique values

```
print(paste0('There are ', n_distinct(lake_joined$event), ' glacier lakes in the dataset'
```

```
[1] "There are 245 glacier lakes in the dataset"
```

That seems a bit high, and if you open up the table youd notice that the `event` column has repeat rows for lakes at different sites. This is good! However for seeing the number of lakes and comparing total size, we should have some way to aggregate the lakes. I will separate!

```
lake_joined <- lake_joined %>%
  separate(event, into = c("lake", NA), sep = "_") # this has the added benefit of renami
```

What's it now?

```
print(paste0('There are ', n_distinct(lake_joined$lake), ' glacier lakes in the dataset')
```

```
[1] "There are 91 glacier lakes in the dataset"
```

91 is a lot to plot but Now I want to examine 4 things graphically:

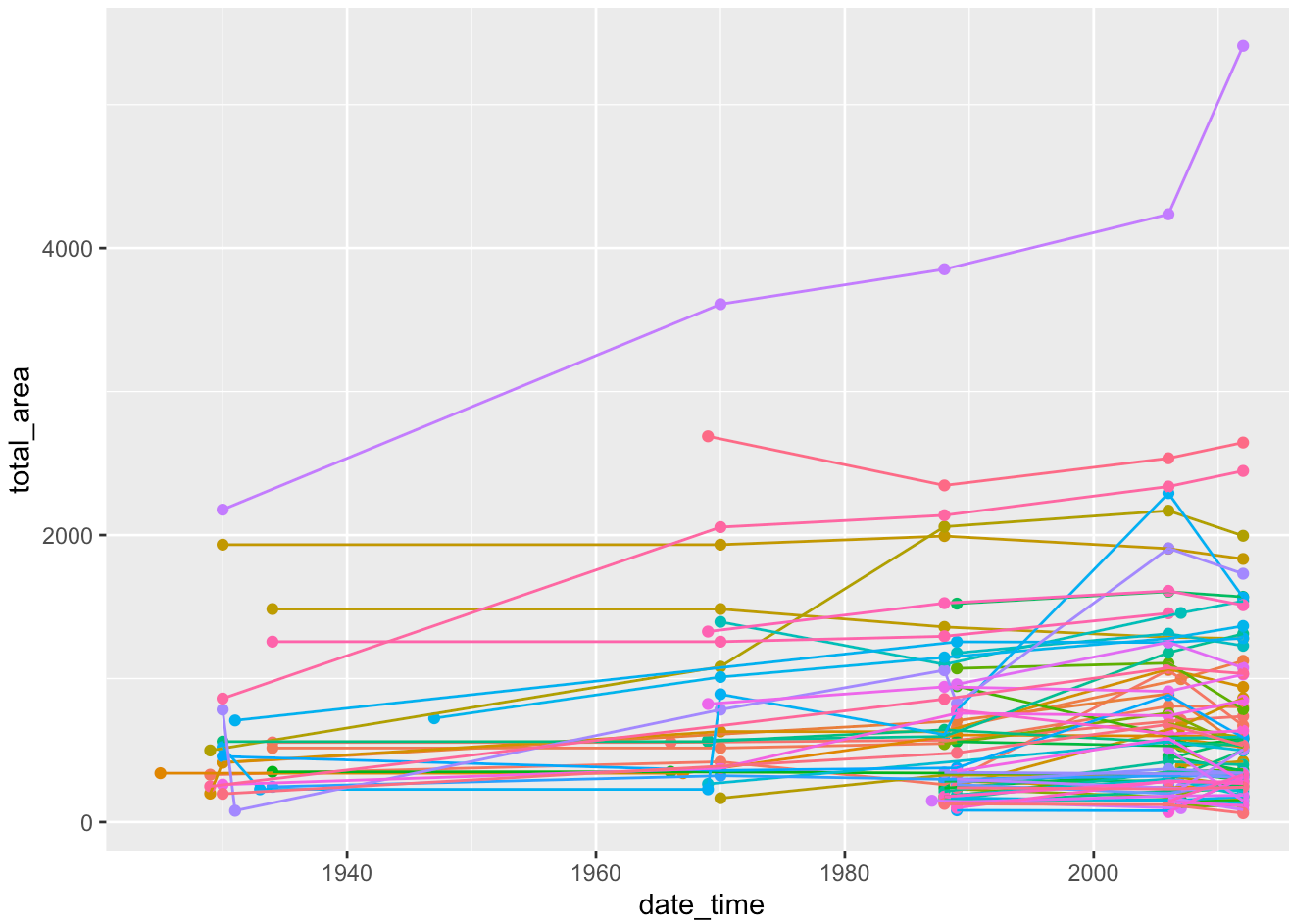
1. How do the glacier lake areas change over time
2. Where are the glacier lakes
3. Where are the glaciers
4. Where do the lakes intersect with the glaciers (aka should I drop any)

##1 How do the glacier lake areas change over time

```
# First we aggregate by glacier lake and year and take a sum of the total area
lake_joined %>%
```

```
  group_by(lake, date_time) %>%
  summarise(total_area = sum(perimeter_m)) %>%
  ggplot(aes(x = date_time, y = total_area,
             color = lake)) +
  geom_point()+
  geom_line()+
  theme(legend.position = 'none')
```

```
`summarise()` has grouped output by 'lake'. You can override using the
`.groups` argument.
```



That is unintelligable for now, but I can see some variation.

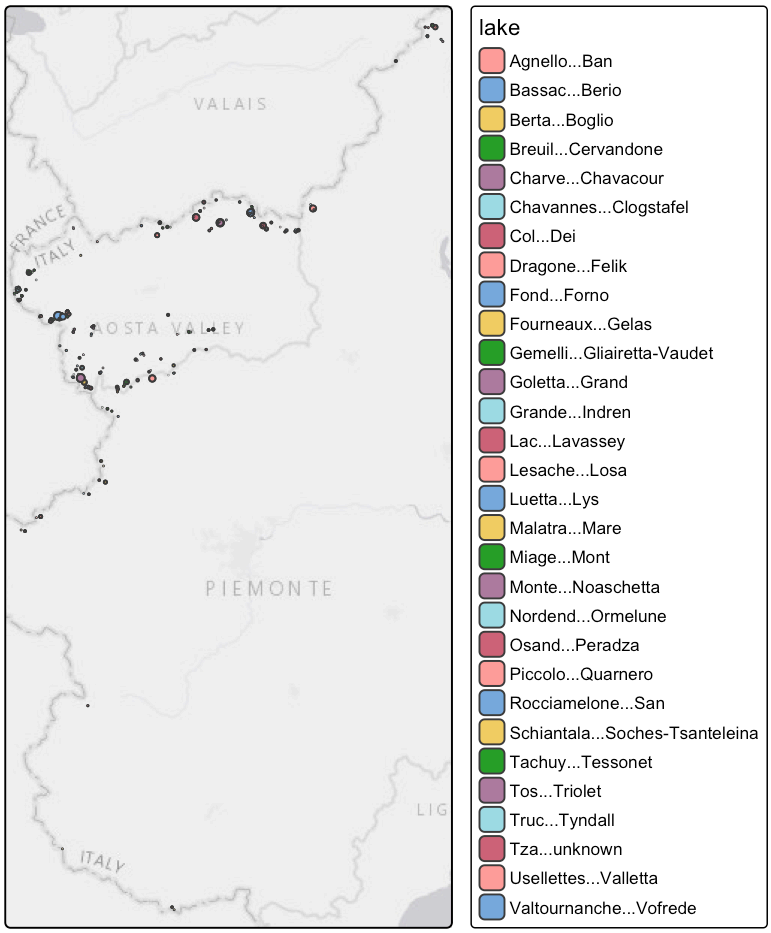
2. Where are the glacier lakes

This is going to require I make the glacier lakes have shape. To do that i'm going to make points at each lake and add a buffer of the outline of the lakes.

```
lake_geo <- lake_joined %>%
  st_as_sf(coords = c("longitude", "latitude"), crs = 4326) %>%
  st_transform( crs = 3857) %>%
  st_buffer( dist = lake_joined$perimeter_m) %>%
  st_transform(crs = 4326)
```

```
# now we can plot


tm_shape(lake_geo)+
  tm_polygons(fill = 'lake')+
tm_basemap()
```

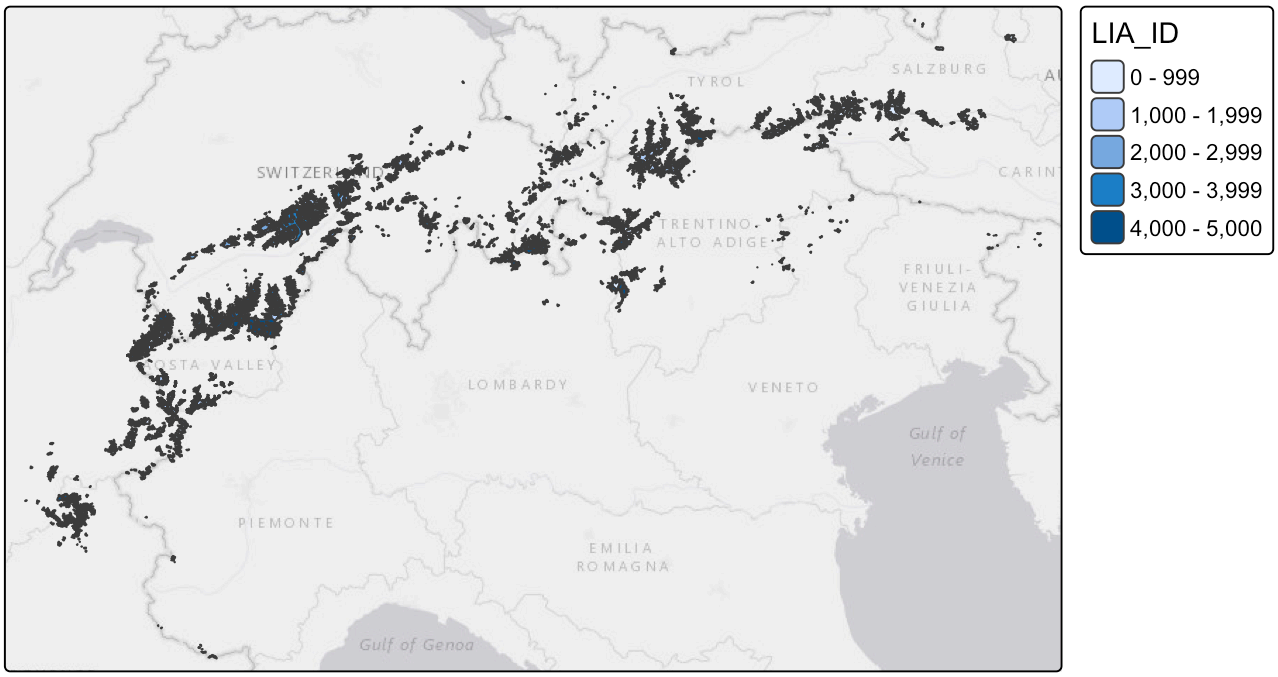
```
Warning: Number of levels of the variable assigned to the aesthetic "fill" of
the layer "polygons" is 91, which is larger than n.max (which is 30), so levels
are combined.

[plot mode] fit legend/component: Some legend items or map compoments do not
fit well, and are therefore rescaled.
ℹ Set the tmap option `component.autoscale = FALSE` to disable rescaling.
```



It looks like our lakes are all over, lets plot out the glaciers and see their extent

3. Where are the glaciers

```
tm_shape(glacier)+
  tm_polygons(fill = 'LIA_ID')+
tm_basemap()
```



I want to crop to the shape of the aosta valley as we have a lot of glaciers and lakes there

```
italy_map <- ne_states(country = "Italy", returnclass = "sf")

aosta_map <- italy_map %>%
  select(province = name, region, geometry) %>%
  filter(region == "Valle d'Aosta")
```

Now lets crop to that region and plot again

```
sf::sf_use_s2(FALSE)
```

```
Spherical geometry (s2) switched off
```

```
glacier <- glacier %>%
  st_transform(crs(aosta_map))

glacier_cropped <- st_crop(glacier, st_bbox(aosta_map))
```

```
although coordinates are longitude/latitude, st_intersection assumes that they
are planar
```

```
Warning: attribute variables are assumed to be spatially constant throughout
all geometries
```

```
lake_geo <- lake_geo %>%
  st_transform(crs(aosta_map))

lake_cropped <- st_crop(lake_geo, st_bbox(aosta_map))
```

```
although coordinates are longitude/latitude, st_intersection assumes that they
are planar

Warning: attribute variables are assumed to be spatially constant throughout
all geometries
```

4. Where do the lakes intersect with the glaciers (aka should I drop any)

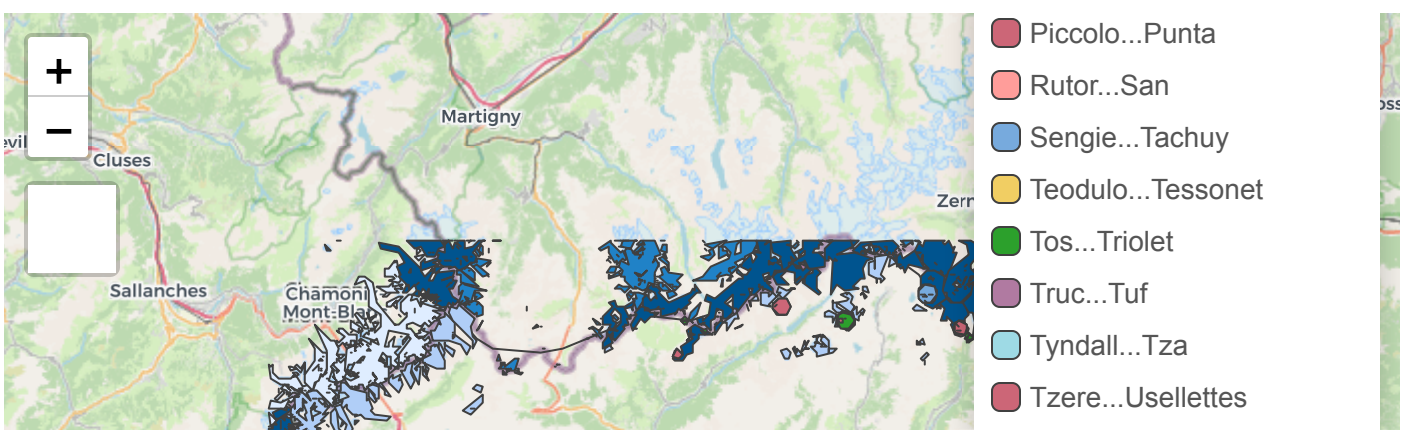
now lets make a big aosta plot

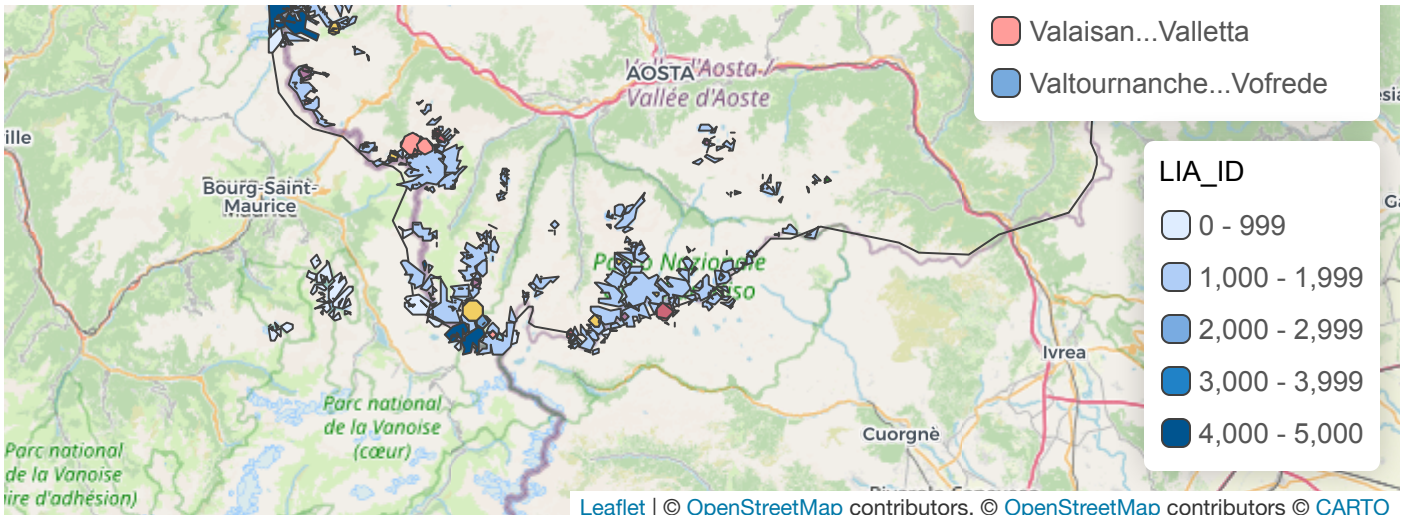
```
tmap_mode("view")
```

```
ℹ tmap modes "plot" - "view"
ℹ toggle with `tmap::ttm()`
```

```
tm_basemap(c(StreetMap = "OpenStreetMap", TopoMap = "OpenTopoMap"))+
   tm_tiles(c(CartoDB = "CartoDB.PositronOnlyLabels")) +
tm_shape(aosta_map)+
  tm_borders()+
tm_shape(glacier_cropped)+
  tm_polygons(fill = 'LIA_ID')+
tm_shape(lake_cropped)+
  tm_polygons(fill = 'lake')
```

```
Warning: Number of levels of the variable assigned to the aesthetic "fill" of
the layer "polygons" is 73, which is larger than n.max (which is 30), so levels
are combined.
```





Questions for fpm 2

1. What have you learned about your data? Have any potentially interesting patterns emerged? Point to specific visualizations that you created as you describe your findings.

I have learned that the extent of my data is much larger than what I thought it would be. I assumed that the glaciers and lakes would naturally all be in Valle D'Aosta, as that is where I have heard of them being. Instead I found that my glacier data spans the entire alps, with my lakes being concentrated in the Valle. However my lakes stretch as far as the bottom of the French alps. My final visualization showed that the area that I expected to look at, in the Northeast of Aosta, only has a couple of lakes.

2. In FPM #1, you outlined some questions that you wanted to answer using these data. Have you made any strides towards answering those questions? If yes, how so? If no, what next steps do you need to take (e.g. I need to create X plot type, I still need to track down Y data, I need to restructure existing data so that you can visualize it in Z ways, etc.)? Have any new questions emerged?

My second question is a YES! I can definitely use glacier lake size to model glacier recession! I have perimeter and it seems to generally get bigger over time. My first question, so far, is a no. I have to do more research into existing data to see if there is mapped historical glaciers.

3. What challenges do you foresee encountering with your data? These can be data wrangling and / or visualization challenges.

This data is big and long spanning. The issue is to make any sort of meaningful visualization I need to lower the scope, maybe to a couple lakes or a smaller area in general. Furthermore I will have to do some further exploration on the actual lakes to see where they all reside, maybe an animation to show all ~9 of them.